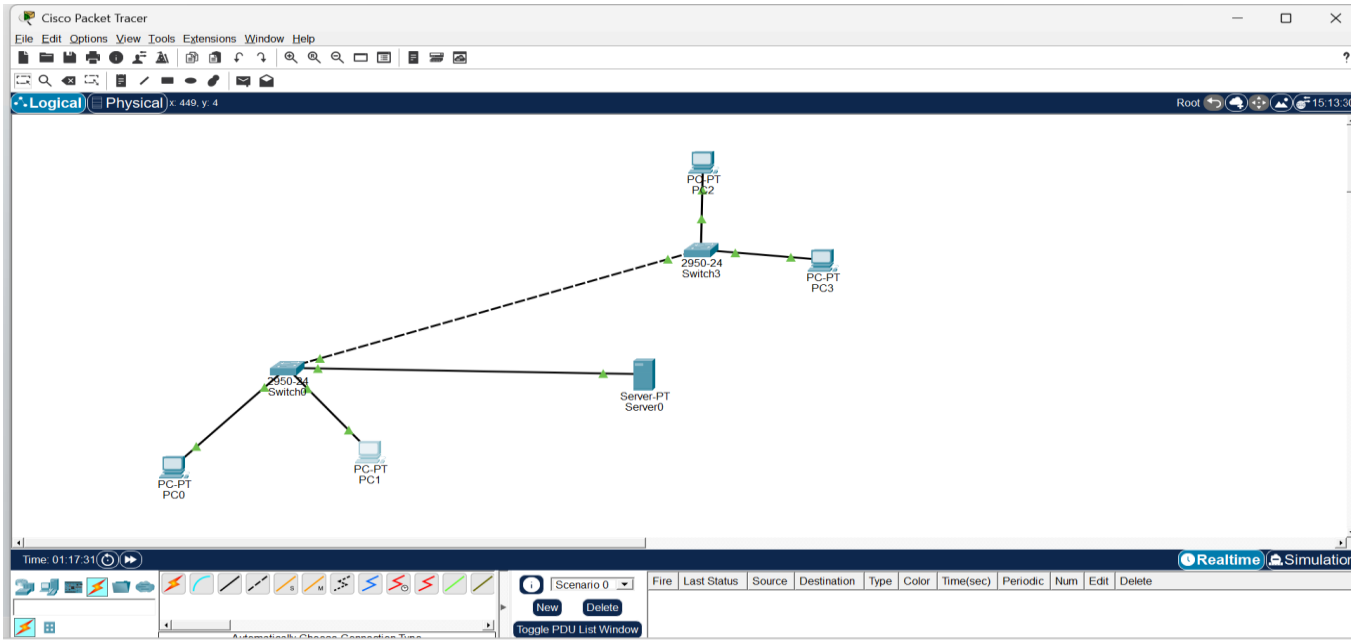
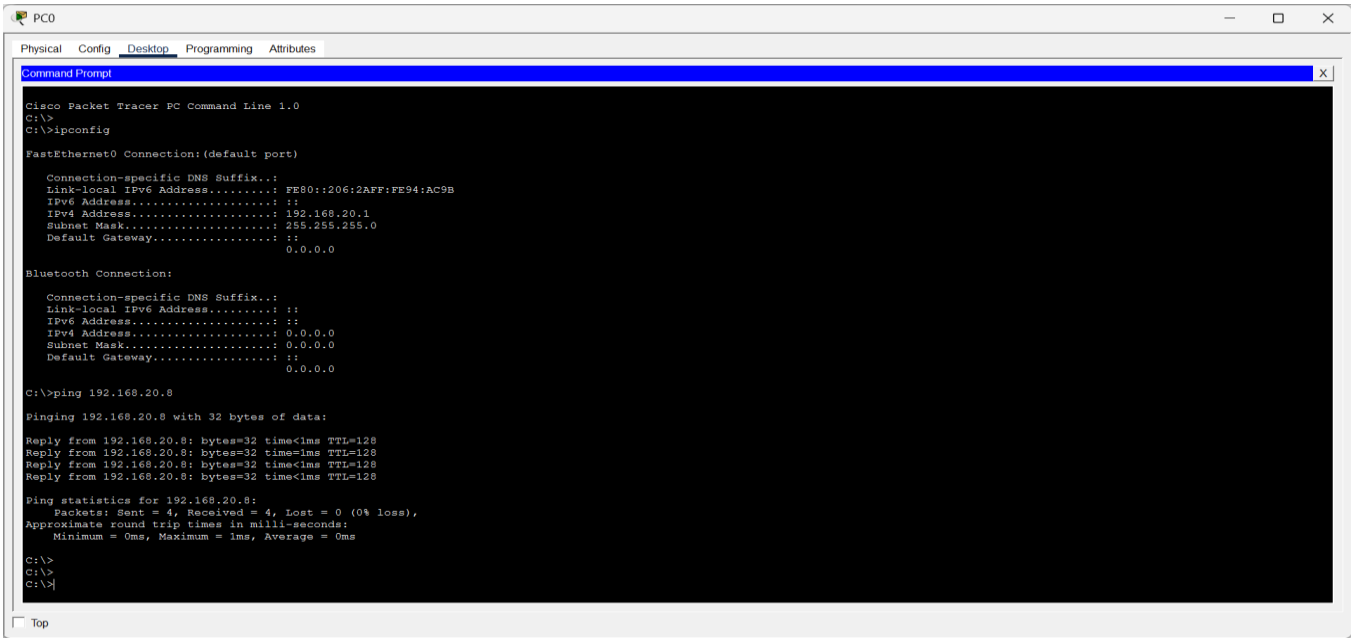
Experiment 3

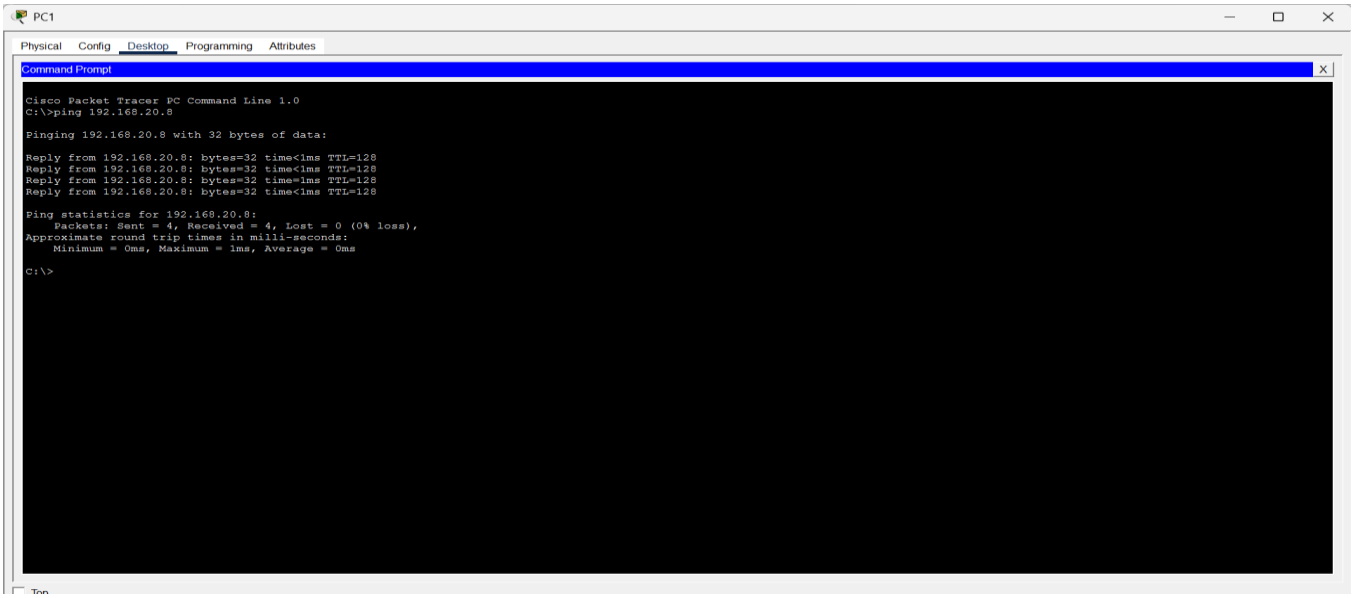
Configuration and Analysis of DNS and DHCP Services for Automated Network Address and Name Management using Cisco Packet Tracer.

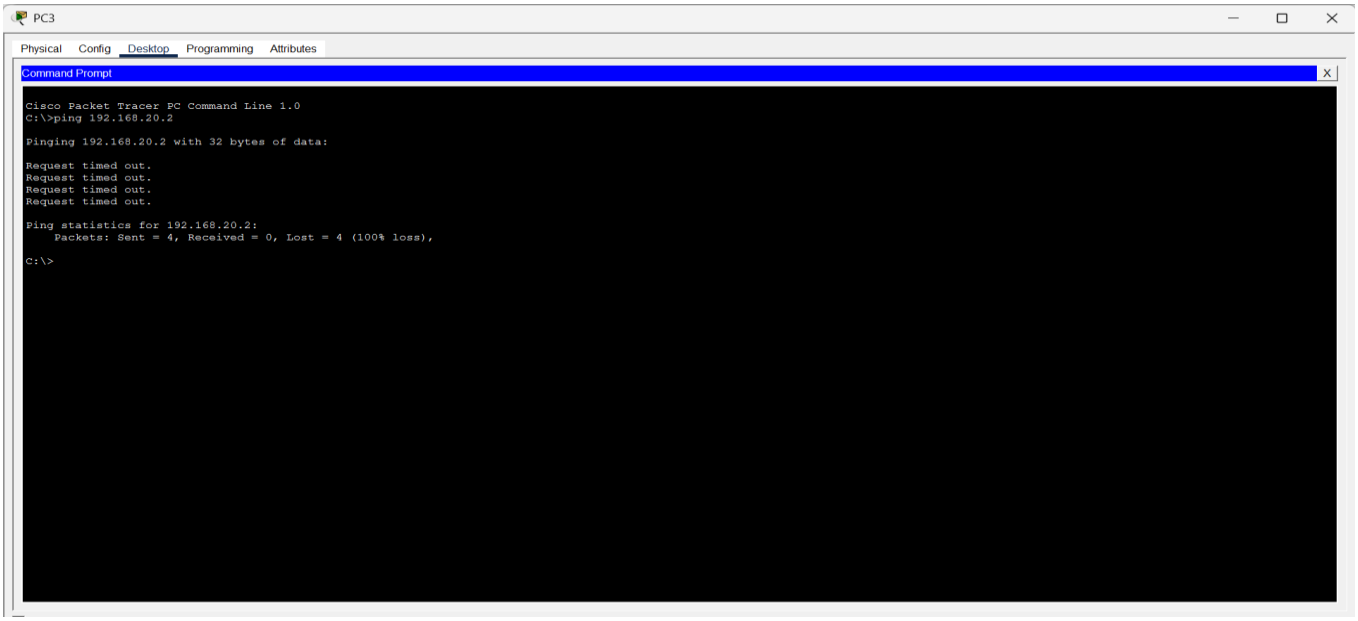
Objectives:

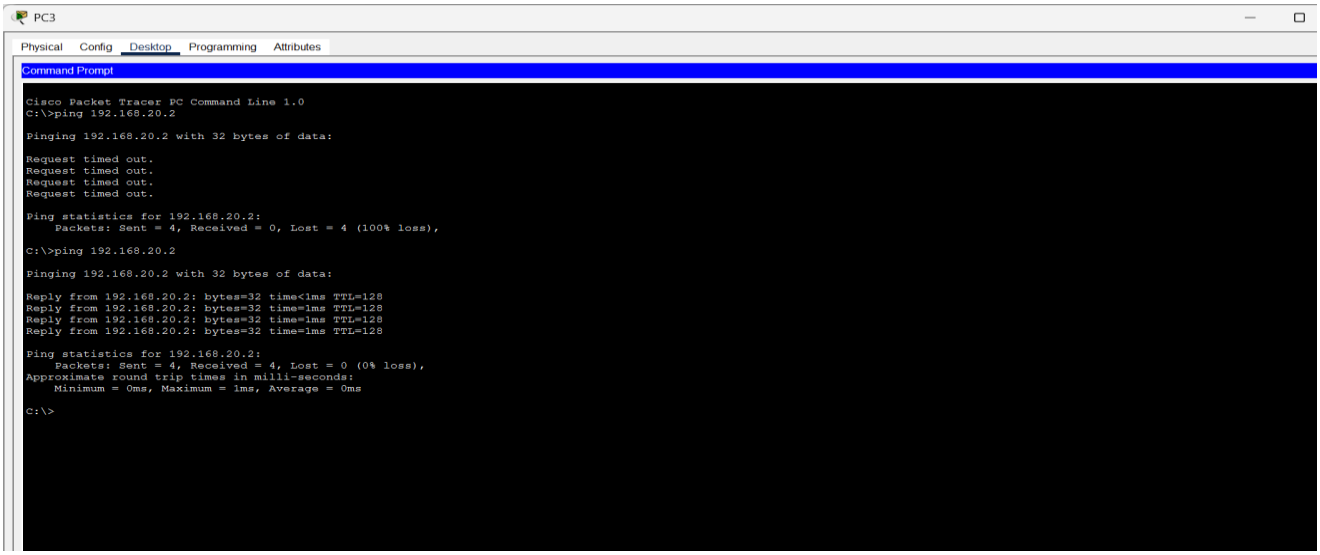
(i) To configure and verify the DHCP protocol for automatic IP address allocation to client devices in a network.






PC1
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.20.8
Pinging 192.168.20.8 with 32 bytes of data:
Reply from 192.168.20.8: bytes=32 time<1ms TTL=128
Reply from 192.168.20.8: bytes=32 time<1ms TTL=128
Reply from 192.168.20.8: bytes=32 time=1ms TTL=128
Reply from 192.168.20.8: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.20.8:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>


PC3
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.20.2
Pinging 192.168.20.2 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Ping statistics for 192.168.20.2:
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\>

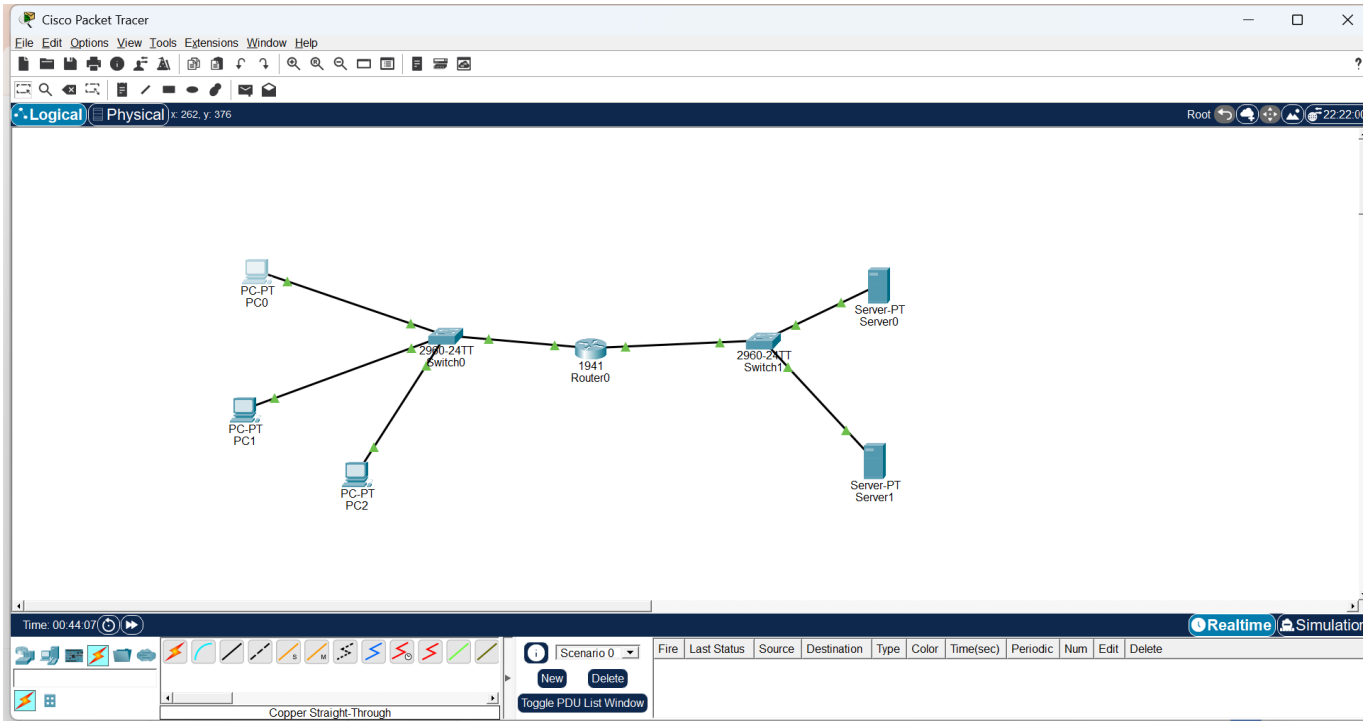

PC3
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.20.2
Pinging 192.168.20.2 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Ping statistics for 192.168.20.2:
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\>ping 192.168.20.2
Pinging 192.168.20.2 with 32 bytes of data:
Reply from 192.168.20.2: bytes=32 time<1ms TTL=128
Reply from 192.168.20.2: bytes=32 time=1ms TTL=128
Reply from 192.168.20.2: bytes=32 time=1ms TTL=128
Reply from 192.168.20.2: bytes=32 time=1ms TTL=128
Ping statistics for 192.168.20.2:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>

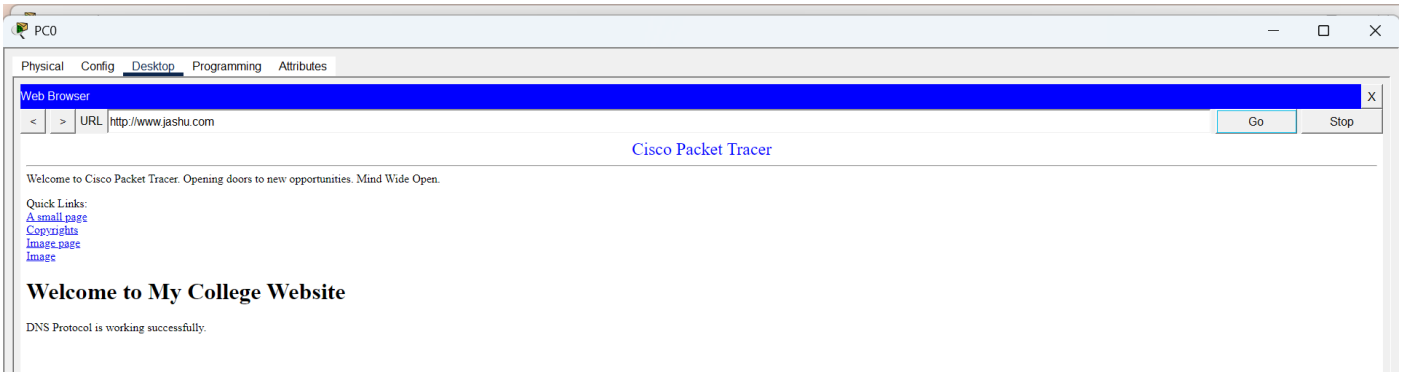
Output:

Objective 1: DHCP protocol

SI. No	Source IP	Destination IP	Data sent (bytes)	Data received (bytes)
1	192.168.20.1	ping 192.168.20.8	4	4
2	192.168.20.2	ping 192.168.20.8	4	4
3	192.168.20.3	ping 192.168.20.2	4	4
4	192.168.20.4	ping 192.168.20.2	4	4
5	192.168.20.3	ping 192.168.20.4	4	4

(ii) To configure and test the DNS protocol for translating domain names into IP addresses and enabling access to network resources using hostnames.




PC0
Physical
Config
Desktop
Programming
Attributes
Web Browser
X
<
>
URL
http://www.jashu.com
Go
Stop
Cisco Packet Tracer
Welcome to Cisco Packet Tracer. Opening doors to new opportunities. Mind Wide Open.
Quick Links:
A small page
Copyrights
Image page
Image
Welcome to My College Website
DNS Protocol is working successfully.